

What remains to finish

PetSafe 100-1339 R03 A | v0.9.9 | 8 October 2026

All 150 catalog entries are drawn on one KiCad sheet. This includes empty options and test pads. The drawing is structurally complete; hidden wiring, candidate identities and component values still need evidence.

Drawing / libraries	Electrical gaps	Packages / values
1 A2 sheet; 20 stock + 4 authorized custom symbol definitions; no missing library files for selected candidates.	28 open pads = 10 populated-entry pads + 18 DNP pads. 0 GPIO pads open.	147/150 footprints; 44 capacitor values and 2 magnetic-part values unknown.

What your via review changed

User confirms V035/RA2/C39.1 to VREF and V042/PIC20/C32.1 to board VDD. Earlier RC2/V049-TP3 and RB0/V026-VREF retained. RC1/V053-V070-R18 confirmed; old RC1/VREF photo assumption explicitly rejected. Also confirmed: RA3-R14, RA4-R15, RA5-R16 and RC0-R11; V053-V071 rejected. 123 IDs reviewed; 120 active sites. PIC15/RC4 now reaches U5.2/INA via V022-V016; PIC16/RC5 reaches U5.3/INB via V023-V019. V011-V014/Q1.L now confirmed to motor VDD/U5 pin4, distinct from logic VDD. R43.2 joins TP14, separate from Q1.L. Four IDs remain unmentioned, some photo-matched. Raw battery BATTERY+ reaches Q1 single pad3; Q1.L/pad2 supplies VSYS. TP14 stays on the raw side (photo). Both are separate from regulated board VDD. R1A supports the FMOS3401A PMOS candidate.

Remaining area	What is missing	How to resolve
PIC / block interfaces	0 GPIO pads open. Other pins have local nets but may lack remote continuations. Motor INA/INB now resolved to PIC15/16.	Review remote continuations of existing local nets; the open-GPIO checklist is complete.
Q8 / RF output / C6	V113/V115 resolve Q8 centre pin5 GND. V114 is U2 pin2 GND; Q8 source supply and C6.1 route remain inferred/unresolved.	V114 conflict resolved. Check Q8 supply independently; no inferred rail merge.
Tuning bank	All five controls mapped: RC6/R13, RA3/R14, RA4/R15, RA5/R16, RC0/R11. Capacitor-midpoint and antenna links remain unknown.	Seven capacitor pairs gave OL. Broad pairwise tests stopped; next trace toward a specific visible/component endpoint.
Auxiliary / receiver	V077 grounds C19/R29. V075 supply plus V064-V070 = 11.2k supports Q7 emitter/VDD, R19 pull-up and R18 base input; topology inferred. V067 joins J3.1/PIR supply; U6A remains DNP. V063 feed now joins regulated VDD; V017/R42.1/R44.1 now joins TP4/PIC25; Q1.L/V011-V014 post-Q1 VSYS confirmed; BATTERY+ is separate; R1A supports FMOS3401A PMOS candidate; R43.2/V015 joins TP14, separate from Q1.L.	RC1 now reaches R18 via V053-V070; no further generic GPIO sweep needed. V071/R21.2 reaches VREF/RB0; other receiver-bias branches and Q7 topology remain inferred. V053-V071 is rejected.
D2 / D6 / candidate parts	D2/4P identity and all three pad routes remain open. D6 has two open pads. V082 reaches logic VDD/TP103; its D3 terminal remains unresolved. U6/Q8 and small BJTs/diodes retain candidate identities/polarities.	D2 diode-mode guide; D6 continuity; targeted candidate pin checks.
Other open fitted pads	C26.1 and TP9.1; U6 NC pins 4/5 have unknown external ties. LED common now grounded from V127/photo; internal LED mapping and S1 legs remain provisional.	Targeted continuity; no need to identify chips on empty footprints.
Component values	44 unmarked fitted capacitor values and L1/L2 type/value remain unrecovered. C5 = 470uF / 16V is recorded; 38 fitted resistors have decoded nominal values.	LCR first on RF/tuning parts, recording frequency; in-circuit results are network readings.
Packages	147/150 footprints assigned. C5, S1 and LED1 still need exact geometry; other photo-based package assignments remain candidates.	C5 diameter/lead spacing; S1/LED1 body and pad spacing. R8/C11 sizes already recorded.

Remaining conflicts for later review

Resolved: V025, V026, V073 and Q8 V113/V115. **V114 resolved:** U2.T2/pin2 GND, not Q8. C5 negative confirmed by V125/V126. V127 grounds the photo-selected LED common pair. **V075:** regulated VDD measured. Q7/R19 terminal selection inferred from IMG_2434 and the new 11.2 kohm V064-V070 result. No functional test supplied.

Validation: KiCad 10.0.5 exports pass; 83 modeled nets match the native file. 36 ERC findings remain: 28 open pins, 3 isolated labels and 5 power findings. This checks the file, not hardware correctness. No artificial NC or power flags were added.

PIC connections: complete picture

Physical package pins, top view. Use the separate numbered close-up to locate them. OPEN means no modeled pad connection; a local connection can still have an unknown onward destination.

Pin	Function	State	Local destination / evidence
1	RE3/MCLR	Local net	J1 VPP; mapping inferred; D6 route missing
2	RA0	Local net	C25 signal pad / V031; onward destination unknown
3	RA1	Local net	R28 + R29 + TP7; user meter / visible trace
4	RA2	Local net	VREF TP via V035; user confirms via-to-TP; C39.1 local photo route
5	RA3	Local net	V036-V108 to R14; user continuity report
6	RA4	Local net	V037-V102 to R15; user continuity report
7	RA5	Local net	V039-V097 to R16; user continuity report
8	VSS	Local net	GND / V041 photo anchor
9	RA7/OSC1	Local net	Y1 oscillator; photo-derived
10	RA6/OSC2	Local net	Y1 oscillator; photo-derived
11	RC0	Local net	V047-V091 to R11; user continuity report
12	RC1	Local net	R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected
13	RC2	Local net	TP3 via V049; user-confirmed; existing R10 branch photo-derived
14	RC3	Local net	TP11; visible route; onward destination unknown
15	RC4	Local net	V022-V016 to U5 pin2 / INA; user-confirmed
16	RC5	Local net	V023-V019 to U5 pin3 / INB; user-confirmed
17	RC6	Local net	R13; user meter / visible trace
18	RC7	Local net	TP17; visible route; old R13 guess withdrawn
19	VSS	Local net	GND / C32 / V045 photo anchor
20	VDD	Local net	V042 to board VDD TP user-confirmed; C32.1 local photo route
21	RB0	Local net	VREF (TP104), V026 and V071/R21.2; user-confirmed
22	RB1	Local net	TP2 / R31; user meter / visible trace
23	RB2	Local net	TP1 / R30; user meter / visible trace
24	RB3	Local net	R4; photo-derived; sensing role remains inferred
25	RB4	Local net	TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived
26	RB5	Local net	TP10 / R35; user meter / visible trace
27	RB6/CLK	Local net	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	Local net	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries (10): C26.1, C6.1, D2.L, D2.R, D2.S, D6.1, D6.2, TP9.1, U6.L1, U6.L2.

Empty options (18): C42.1, C42.2, C49.1, C49.2, J6.1, J6.2, Q9.S, R36.1, R36.2, R46.1, R46.2, R47.1, R47.2, U6A.L, U6A.R2, U7.1, U7.2, U7.3.

Pad names above are photo-model names; U6.L1/L2 map to its internal NC pins 4/5. DNP pads do not block understanding of the fitted circuit. Vias with a component association do not automatically prove a connection to every other pad on a previously inferred net.

Evidence: user via review and supplied MX512H diagram; original front/rear photographs; reconstruction.json, via_audit.json, v09_net_changes.json, v091_net_changes.json, v092_net_changes.json, v093_net_changes.json, v094_net_changes.json, v095_net_changes.json, v096_net_changes.json, v097_net_changes.json, v098_net_changes.json, v099_net_changes.json and native KiCad validation. Pin names follow the existing Microchip datasheet audit. The via viewer retains reports and conflicts.